Self-assembled multilayers and enhanced superconductivity in $(YBa_2Cu_3O_{7-x})_{0.5}:(BaZrO_3)_{0.5}$ nanocomposite films

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Nanocomposite $(YBa_2Cu_3O_{7-x})_{0.5}:(BaZrO_3)_{0.5}$ thin films were fabricated on (001) oriented $SrTiO_3$ substrates by pulsed laser deposition using a single uniformly mixed target. Both x-ray diffraction and transmission electron microscopy revealed remarkable, spontaneous formation of $YBa_2Cu_3O_{7-x}$ (YBCO) and $BaZrO_3$ (BZO) multilayers. The high integrity and continuity of the multilayer made it possible to achieve a critical temperature of 88 K, given that the $BaZrO_3$ fraction in the films is 50 mol %. The unique self-assembled microstructure led to a surprising field dependent critical current density along the *ab* plane. © *2009 American Institute of Physics*. [doi:10.1063/1.3257238]

I. INTRODUCTION

It is known that power applications of superconductors require an effective immobilization of vortices or in other words a high pinning force.[1] The pinning force arises from the presence of localized defects or crystalline imperfections. It has been experimentally shown that nanoscale defects within $YBa_2Cu_3O_{7-x}$ (YBCO) materials can pin the magnetic flux lines, which translates to a high critical current density (J_c). Recent efforts have focused on nonsuperconducting additions to YBCO films to enhance the current carrying capability in applied magnetic fields. For example, Y_2O_3, CeO_2, YBa_2CuO_5, and $BaZrO_3$ (BZO) because of their good compatibility with YBCO, have been effectively used to increase J_c in YBCO films.[2–7] In general, the increase in the BZO content (or the density of pinning sites) initially increases the macroscopic J_c, but beyond some optimum density the performance deteriorates.[1] For this purpose, the critical temperature (T_c) and the J_c of YBCO film with different levels of BZO inclusion have been widely investigated. It is a common view that the T_c decreases linearly with increasing BZO concentration.[8–10] Currently, the reported maximum inclusion of BZO in YBCO films is 8 vol % (~18 mol %) where a T_c of 82.8 K was observed.[8] For this concentration, BZO self-assembles in vertical columns along the *c*-axis.[7,8] A question that arises is, what will happen to the structure and the superconductivity of a YBCO film when the BZO inclusion is more than 18 mol %? In the present work, we investigated the superconductivity and microstructure of $(YBCO)_{0.5}:(BZO)_{0.5}$ films. A high T_c of 88 K and a surprising field dependent of J_c along the *ab* plane were observed, which can be attributed to the self-assembled multilayer structure.

II. EXPERIMENTS

Nanocomposite $(YBCO)_{0.5}:(BZO)_{0.5}$ (YBCO:BZO) films were grown on (001) oriented $SrTiO_3$ (STO) substrates by pulsed laser deposition (PLD) using a XeCl excimer laser ($\lambda = 308$ nm). A single ceramic pellet with a molar ratio of YBCO/BZO=1:1 was used as the target. A substrate temperature of 820 °C and an oxygen pressure of 200 mTorr were maintained during the deposition. Following the deposition, the films were cooled down to room temperature in an oxygen pressure of 300 Torr without any further thermal treatment. The thickness of films measured by the cross-sectional transmission electron microscopy (TEM) was 150 nm. The crystal structure was investigated by x-ray diffraction (XRD) using a Siemens D5000 four-circle diffractometer. The critical temperature (T_c) was evaluated by both alternating current (ac) susceptibility and transport measurement techniques. The critical current density (J_c) was obtained by the standard four-point method from a bridge using the 1 μV/cm criterion, where the films were patterned using standard photolithography with a bridge width of ~250 μm. The angular and field dependent J_c measurements were conducted in a magnetic field (H) rotated in the plane perpendicular to the plane of the film but always normal to the current ($J \perp H$, maximum Lorentz force configuration). With the sample immersed in liquid nitrogen the measurement temperature was varied from 65 to 75.5 K.

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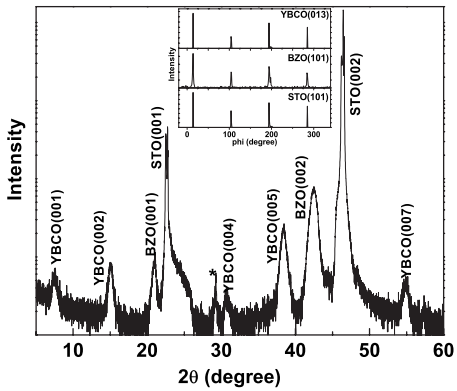

FIG. 1. Typical XRD θ-2θ scan of YBCO:BZO films. A weak BZO (101) peak is denoted by a star. The inset shows phi-scans on STO (101), BZO (101), and YBCO (013) reflections.

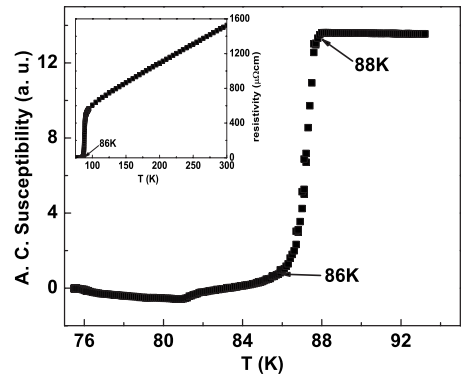

FIG. 2. ac susceptibility of a nanocomposite YBCO:BZO film. The inset shows the resistivity vs temperature characteristic of the same film.

III. RESULTS AND DISCUSSION

Figure 1 shows a typical XRD θ-2θ scan of the YBCO:BZO films. As can be seen from the figure, the YBCO and BZO phases are well defined and have comparable intensities, as broadly expected for the volume fractions and atomic weights of the compounds. The YBCO shows only (00*l*) diffraction peaks and hence is preferentially oriented along the *c*-axis (perpendicular to the substrate surface). The *c* lattice constant of YBCO, calculated from the (005) peak, is 11.66(96) Å, which is close to the bulk value of 11.6805 Å (see below for further discussion). The BZO shows (001) and (002) peaks, indicating the same *c*-axis orientation. It should be pointed out that besides the (00*l*) peaks of YBCO and BZO, there is a weak peak at around $2\theta=29.4°$ (shown with a star in Fig. 1), which can be attributed to BZO (101). By comparing the relative intensities of the Bragg peaks associated with the BZO (002) and BZO (101) peaks we estimate that ~97% of the BZO is *c*-axis oriented. The full-width at half-maximum of the rocking curves for YBCO (005) and BZO (002) are 1.24° and 0.86°, respectively (not shown). These values are higher than those for typical pure YBCO and BZO films and the large lattice mismatch between YBCO and BZO is likely responsible for this. The in-plane orientation of the YBCO:BZO films with respect to the major axis of the STO substrate is revealed by phi-scans on the YBCO (013), BZO (101), and STO (101) reflections (shown in the inset of Fig. 1). The orientation relationship is determined to be $(001)_{YBCO}\|(001)_{BZO}\|(001)_{STO}$ and $[010]_{YBCO}\|[010]_{BZO}\|[010]_{STO}$, which is in accordance with the pure YBCO and BZO thin films grown on STO substrates.

The temperature dependence of ac susceptibility of YBCO:BZO film is shown in Fig. 2. A clear superconducting transition was observed and the on-set T_c was determined to be 88 K, which is much higher than the reported value of YBCO films with only 18 mol % BZO.[8,9] To confirm the high T_c, the temperature dependence of resistivity measurement was conducted and shown as inset of Fig. 2. A typical linear relationship was found above the transition temperature. The zero resistance temperature of 86 K from the transport measurement agrees well with the result obtained from the ac susceptibility measurement. The resistivity values of the YBCO:BZO film, 600 and 1500 $\mu\Omega$ cm at 100 and 300 K, respectively, are one order of magnitude higher than those of a typical pure YBCO film. The layered structure and the highly resistive nature of the BZO lead to the high resistivity of the nanocomposite films (see below for further discussion).

It is very important to investigate the origin of the unexpected high T_c in the present study, given that the BZO fraction in the films is 50 mol %, whereas in standard YBCO +BZO films with lower BZO concentrations, T_c decreases much more rapidly (~0.31 K/mol %).[8–10] It is well known that there is a close relationship between the superconductivity and the microstructure in YBCO films.[1] In order to reveal the microstructure of the YBCO:BZO films, low magnification and high resolution cross-sectional TEM were performed. As can be seen from the low magnification bright-field TEM image (Fig. 3), a multilayer structure is formed throughout the film, i.e., spontaneously ordered alternating layers of YBCO and BZO. The YBCO and BZO layers show thicknesses of 6–10 and 7–11 nm, respectively. All the layers are continuous in-plane (lateral dimension) up to a micrometer scale. This is the first time a self-assembled, ordered horizontal nanocomposite film has been observed, although we have observed a self-assembled vertical analog before in another perovskite system.[11,12] For YBCO films with BZO, the microstructures previously observed are either random BZO

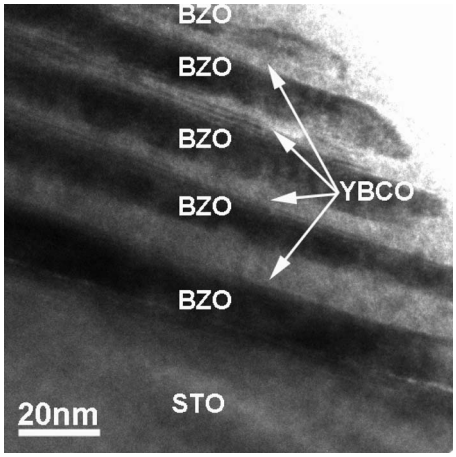

FIG. 3. A low magnification bright field TEM image of a nanocomposite YBCO:BZO film on STO substrate.

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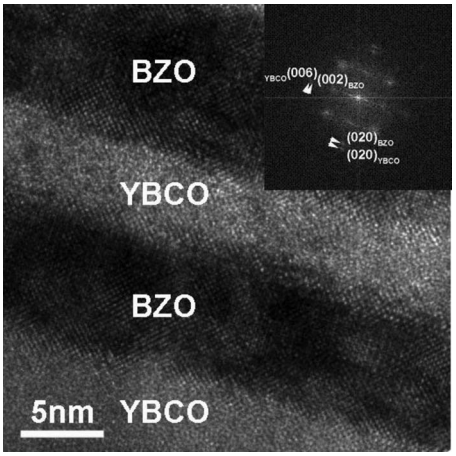

FIG. 4. HRTEM image with corresponding FFT filtered image at the YBCO/BZO interface.

nanoparticles or self-assembled columnar BZO nanorods.[5–7,13] The completely different microstructure in the present study originates from an interplay between composition, strain, and growth kinetics.[11] A high resolution TEM (HRTEM) image (Fig. 4) reveals excellent heteroepitaxial growth between the YBCO and BZO layers. The corresponding fast Fourier transform (FFT) image from the interface of YBCO and BZO is shown in the inset of Fig. 4. The orientation relations of the YBCO/BZO multilayers are determined to be $(001)_{YBCO} \| (001)_{BZO}$ and $[010]_{YBCO} \| [010]_{BZO}$, which are consistent with the results from XRD measurements. This unique and micron scaled-continuous self-assembled nanolayer structure is believed to be the main reason for the high T_c value.

Considering the multilayer structure in the YBCO:BZO films, a T_c of 88 K is not completely unexpected. Huhtinen *et al.*[14] represented a systematic investigation of superconductivity of YBCO/BZO multilayers fabricated from two different targets by PLD. T_c from 88 to 89.3 K have been observed, which dependent on the thickness of BZO layer relative to the YBCO layer and agree well with our results. It should be pointed out that multilayers have been found to increase the density of misfit dislocations at the layer interfaces, which compensate the mismatch between YBCO and BZO lattices in the current study.[14–17] This is the main reason that the c lattice constant of YBCO phase is close to the bulk value, which also contributes to the high T_c value.[14]

To investigate the flux pinning properties of the YBCO:BZO films, the magnetic field (H) dependence of J_c was measured at 65 K and shown in Fig. 5. J_c is normalized to self-field value $[J_c(sf)=0.15\ \mathrm{MA/cm^2}]$ to facilitate the comparison. First, a most striking feature is observed for the case of $H \| ab$, which has an extremely slow J_c field decay. The α value in the relation of $J_c \sim H^{-\alpha}$ was calculated to be 0.06, which is much lower than the typical value of pure YBCO film ($\sim$0.51) and indicates a weaker field dependence of J_c. Second, for $H \| c$ $J_c(H)$ drops much faster than standard pure YBCO with a α value of 1.48. For the lower concentration doping of BZO in YBCO films, BZO have been reported to self-assemble in vertical columns along the c-axis. Moreover, α values from 0.31 to 0.61 have been observed.[7,8]

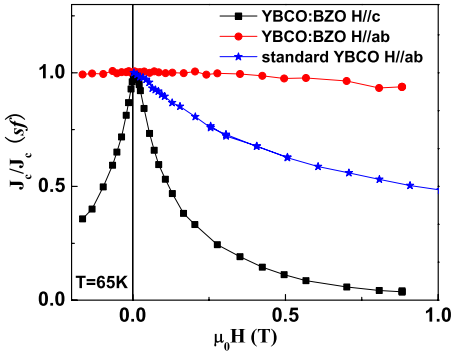

FIG. 5. (Color online) Magnetic field dependence of J_c of a nanocomposite YBCO:BZO film at 65 K. For comparison, a curve for standard pure YBCO films is also included.

By comparing these values with those in the current study, we can see the much lower density of defect along c-axis and much higher density of defect along ab plane in the nanocomposite YBCO:BZO film, which is confirmed by the multilayer structure. It should be pointed out that the $J_c(sf)$ value here is comparable to the reported results of YBCO/BZO multilayers fabricated by two different targets using PLD and could be related to the high resistivity observed.[14] The angular dependence of J_c was also measured and shown as Fig. 6. An enhanced peak for $H \| ab$ was observed compared with the typical pure YBCO films and no peak was found for $H \| c$, which are consistent with the multilayer structure and the reported results.[10,15,18]

IV. CONCLUSIONS

In conclusion, nanocomposite YBCO:BZO thin films with a BZO inclusion of 50 mol % were fabricated by PLD. Both YBCO and BZO phases exhibited spontaneous, horizontal, multilayered, and heteroepitaxial phase assemblage. This unique microstructure led to a T_c of 88 K and a surprising field dependence of J_c along the ab plane. The present study represents a new route to prepare multilayer structures of the superconducting materials and holds great promise for other functional oxide systems too.

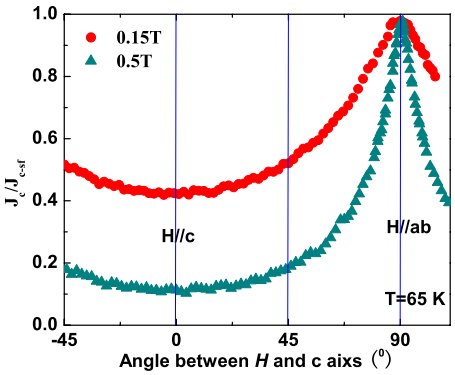

FIG. 6. (Color online) The angular dependence of J_c of a nanocomposite YBCO:BZO film at 65 K and with applied magnetic field of 0.15 and 0.5 T. In the figure, 0° corresponds to $H \| c$ and 90° corresponds to $H \| ab$. J_c is normalized to $J_c(sf)$ to facilitate the comparison.

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